

Executive Summary

PASS RATE 35% target 90%	TOTAL TESTS 168 executed	PASSED 59 35% of total	FAILED 109 65% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 35% | Target: 90% | Gap: +55%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 35% overall pass rate across 168 tests, substantially below the 90% target threshold. The API Automation suite shows particularly low reliability at 33.75% pass rate (27/80), while Backend Smoke Validation performs marginally better at 36.36% (32/88), indicating widespread endpoint instability and response inconsistencies throughout the system. The high failure rate suggests systemic issues within the backend service layer that require immediate investigation and remediation to achieve production readiness. These results indicate the backend APIs are not meeting expected behavioral specifications and require comprehensive debugging and stabilization efforts.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	27	49	36%	FAIL
2	BE Smoke Validation	88	32	56	36%	FAIL
TOTAL		168	59	109	35%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#44	03 Jul	168	35%	109	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	59	35.1%	180ms	Within SLA
Requests Failed	109	64.9%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	6%
High	8	7%
Medium	4	4%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the 105 failing endpoints to identify common failure patterns	Immediate	P1
2	Prioritize debugging of the 53 failing API automation endpoints for service stability improvements	3 days	P1
3	Review backend service logs and error responses to categorize failure types (timeout, 5xx errors, malformed responses)	5 days	P2
4	Implement enhanced error handling and response validation in the backend services	1 week	P2
5	Deploy incremental fixes starting with smoke test failures to establish baseline stability	1 week	P2
6	Schedule technical review session with backend development team to address architectural concerns affecting endpoint reliability	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints demonstrate 65% failure rate indicating widespread service instability
2	API Automation suite shows critical reliability issues with only 27 of 80 endpoints responding correctly
3	Smoke validation failures suggest fundamental backend service configuration or deployment problems
4	Current system performance at 35% pass rate is 55 percentage points below the 90% operational target
5	Both test suites show consistent failure patterns pointing to systemic backend architectural issues

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		36%
BE Smoke Validation		36%